

Bio

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Peng Zhang

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- Specializes in adaptive trial design and statistical monitoring of clinical trials
- Leads internal development of R packages and R Shiny applications to support regulatory and data review workflows

Previous Related Presentations

- **RISW 2025:** *Ensure the statistical validity in context of open-source approach* – Peng Zhang, Christine Matakovich, Peilin Zhou, Tai Xie
- **BBSW 2025:** *Ensure the Statistical Validity in the Context of Open-source and Large Language Model (LLM)* – Peng Zhang, Tai Xie, Christine Matakovich, Peilin Zhou



Challenges for Small- to Mid-Size Organizations to Build a GxP-Compliant R Environment

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Agenda

- 01** Background
- 02** Challenges
- 03** System Level Considerations
- 04** Package Management
- 05** Good Development Practice in R
- 06** Potential AI Usage

Background

Increasing R Usage in Pharma Industry

Automated Report + Interactive Dashboard + LLM Integration

Great Open-Source Community

Publicly available resources from Pharmaverse, R Consortium, R Validation Hub, CAMIS, Openstatsware

Acceptable Submissions to Regulatory

Pilot studies from large pharma companies and working group demonstrates the feasibility

Challenges

1

No Validated Infrastructure

Lack of validated environments to run R for regulated work

2

Open-Source Talent Gap

Few staff with hands-on open-source R expertise

3

Package & Risk Management

Unclear how to manage packages, versions, and associated risks

4

GxP Compliance Knowledge

Limited understanding of what GxP compliance requires

5

Statistical Differences: SAS vs R

Uncertainty about numerical equivalence between SAS and R

6

Regulatory Submission Process

Unclear path to submit R-based analyses to regulators

System level Considerations



Role-Based Access Control

Organization and project-level access control of folders, outputs, and Shiny Apps.



Audit Trails

Captures user activity and system changes.



Validated Infrastructure

Supports reproducibility and traceability.



R Version(s)

Ensures consistency across all data scientists.



Pre-installed Dependencies

Supports usage of R Packages and R Shiny.



Integration with External Systems

Connects with database and file management systems.

Package Management

01 Risk Assessment

Evaluate package quality & fit

- ▶ Use only well-maintained, high-quality packages
- ▶ Match packages to business needs (e.g. {mmrm} for modeling)
- ▶ Approach by R Validation Hub White Paper [1]



02 Qualification

Validate before deployment

- ▶ Confirm package builds in the org repository
- ▶ Verify installation succeeds end-to-end
- ▶ Validate usability with system dependencies



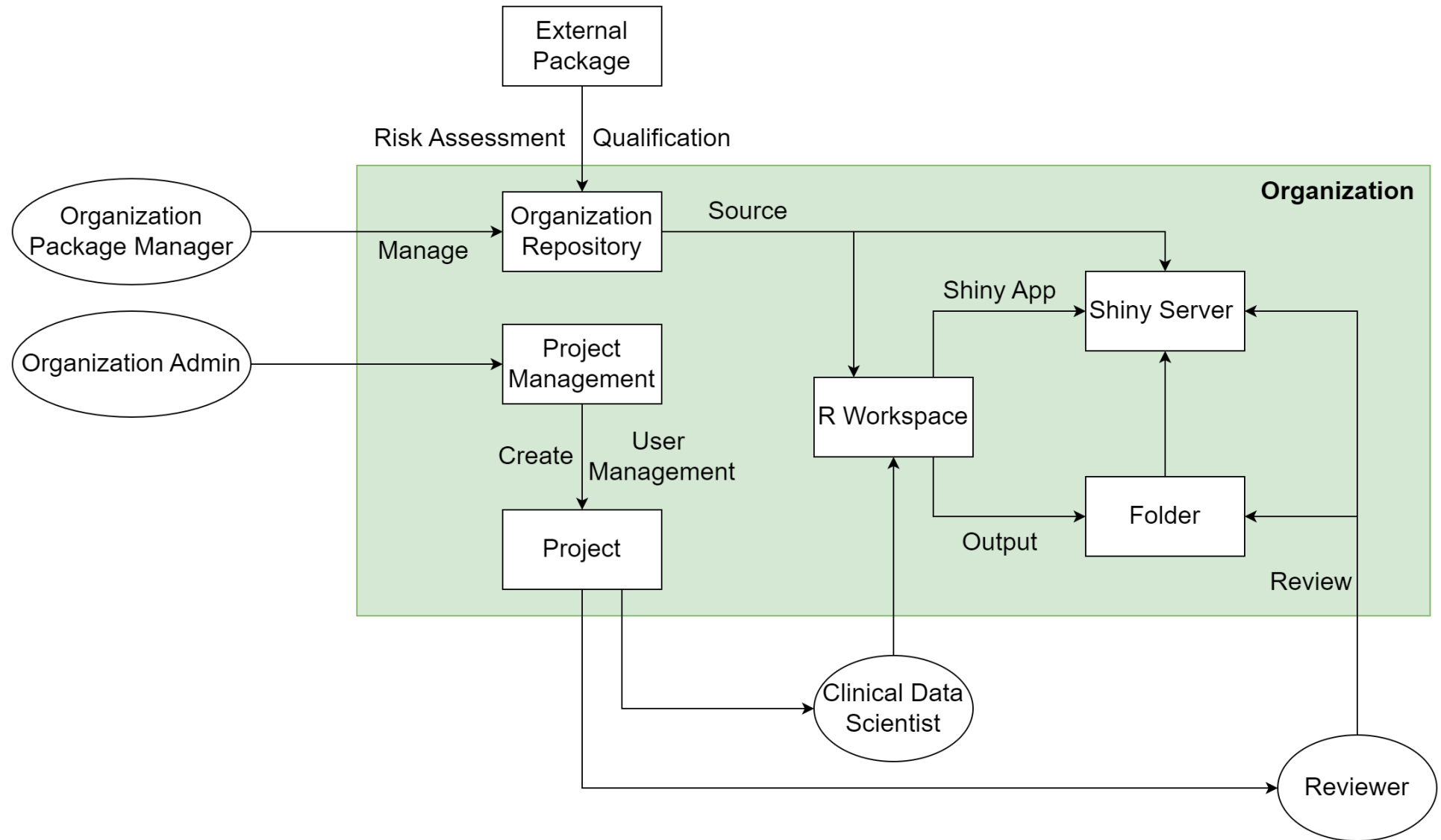
03 Versioned Repository

Pre-installed & version-locked

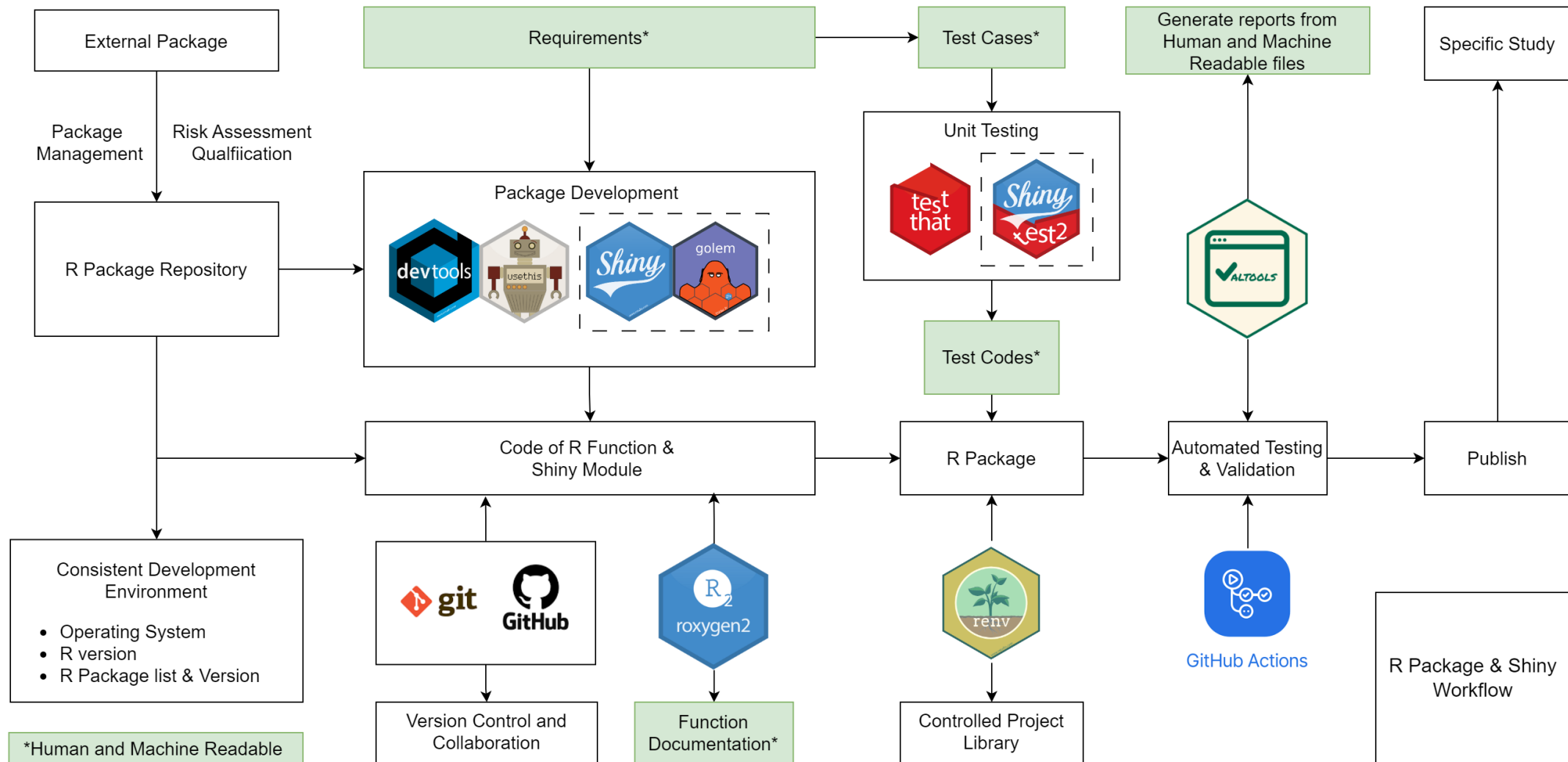
- ▶ Library available to users out of the box
- ▶ Consistent package & dependency versions

[1] <https://pharmar.org/risk/>

Illustrative R Workflow: Roles & Components



Good Development Practices



Potential AI Usage

Organization's AI Policy

Align with internal AI governance and approval workflows

Collaborate with IT + QA

Engage IT and QA early on tooling, access, and validation

PHI / PII & Regulatory

HIPAA, GDPR considerations; confirm if a BAA applies

Data Retention & Training

Synthetic data as an alternative for model training

Confidential Info & IP

Protect trade secrets and proprietary information

Audit Logs

Maintain traceable logs of AI interactions

Conclusion

A GxP-compliant R environment is achievable

You don't need to be a big pharma — the building blocks are accessible to small and mid-size teams.

Start with infrastructure and governance

Prioritize a validated environment, role-based access, and a versioned package repository before scaling use.

Leverage the open-source community

Pharmaverse, R Validation Hub, R Consortium, CAMIS, and Openstatsware reduce what you have to build alone.

Integrate LLMs deliberately, not casually

Define policies, protect data, log interactions — controlled use unlocks productivity without compliance risk.

Thank you!

We welcome any questions or thoughts.